

YUHANG JIANG

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[Homepage](#) | [Google Scholar](#) | [GitHub](#)

Summary

- EIT Manufacturing double-degree master's in Computer Science (UNITN) and Data Science (SUPSI).
- First author of five 2026 papers (three sole-authored) across embodied AI, mechanistic interpretability of state-space models, adversarial robustness, and remote sensing.
- Research intern at Huawei Pisa Research Center and FBK.
- Kaggle Competitions Expert.

Education

EIT Manufacturing Master & Doctoral School (Double Degree) Sep 2023 - Mar 2026

Data Science and AI for a Competitive Manufacturing

- **Università di Trento (UNITN), Italy**
MSc, Computer Science · GPA 98/110
- **University of Applied Sciences and Arts of Southern Switzerland (SUPSI), Switzerland**
Master of Engineering (MSE), Data Science · GPA 5.1/6.0
- **Scholarship:** Recipient of the EIT Manufacturing Scholarship (10,000 EUR/yr)

Anhui University (Project 211), China

Sep 2019 - Jun 2023

BEng, Intelligent Science and Technology

- **Average Score:** 84.34/100
- **Awards:** Second Prize for Outstanding Academic Performance (2022); First Prize Scholarship of Academic Science and Technology (2021)
- **Courses:** Machine Learning, Pattern Recognition, Big Data Analysis, Numerical Analysis, Natural Language Processing.

Publications

"PlnVerify: An Offline Embodied Benchmark for Active Instance Verification"

Poster, FMEA Workshop @ CVPR 2026, [Project Page](#), [arXiv:2605.30639](#), [OpenReview](#)

Master's thesis project.

Proposes Active Instance Verification, a new embodied task. An agent actively selects viewpoints to confirm a fine-grained object description before acting, then recovers on failure. On-device MLLM agents fine-tuned via LoRA and RL.

Yuhang Jiang (sole author)

"GeoSelect: Spatial-Program Execution for Training-Free Referring Remote Sensing Image Segmentation"

IEEE TGRS, under review, [Project Page](#), [arXiv:2607.03869](#)

A training-free neuro-symbolic method that synthesizes and executes typed spatial programs over vision-language models to resolve referring expressions in remote-sensing images. It more than doubles the best prior training-free method on RRSIS-D.

Yuhang Jiang, Guohui Deng*, Miaozhong Xu, Chao Ruan, Jinling Zhao, Linsheng Huang

"Task Structure Reverses Layerwise State Encoding in Sequence Models"

Under review, [arXiv:2606.00926](#)

Shows that layerwise state encoding in sequence models reverses with task structure, via probing and causal intervention across Transformers, Mamba, and RNNs.

Yuhang Jiang (sole author)

"Detection vs. Execution: Single-Bucket Probes Miss Half the Mamba-2 State Sink"

Under review, [arXiv:2606.00930](#)

Shows single-bucket probes miss much of the Mamba-2 state sink's causal substrate: representational similarity does not imply functional equivalence.

Yuhang Jiang (sole author)

"The Scissors Effect: When Resize-Based Input Diversity Helps or Hurts Transfer Attacks"

Under review, [arXiv:2606.22516](#)

Resize-based input diversity helps transfer attacks from standard surrogates but hurts robust ones, explained through gradient geometry.

Yuhang Jiang, Xiaojing Chen

"CL-RAG: A Closed-Loop Multimodal Retrieval-Augmented Generation Architecture for Robust Human-Robot Control Interaction"

WRC SARA 2025 (oral)

Bowen Zhang, **Yuhang Jiang**, Lingxiang Hu, Dun Li, Qianqian Hu*

"Improved lightweight identification of agricultural diseases based on MobileNetV3"

CAIBDA 2022 (oral)

Yuhang Jiang*, Wenping Tong

Software Copyright of Intelligent contract software for agricultural insurance compensation

2022SR1568111.

Note: * indicates the corresponding author.

Academic Service

Reviewer, FMEA Workshop @ CVPR 2026.

Work Experience

Huawei Pisa Research Center

Apr 2026 - Oct 2026

Research Intern

Pisa, Italy

- Working on memory systems for LLM agents, focused on long-horizon, context-persistent behaviour.
- Conducting LLM post-training, applied to automotive AI.

Fondazione Bruno Kessler - FBK

Mar 2025 - Mar 2026

Research Intern

Trentino-Alto Adige, Italy

- Proposed a new embodied perception task and built the first offline multi-view benchmark for active instance verification in realistic 3D environments.
- Designed an MLLM-based agent with attribute decomposition, confidence-weighted multi-view tracking, and next-best-view planning; fine-tuned via LoRA with SFT followed by RL alignment, achieving 85.6% accuracy.

Chengdu Jiaoda Guangmang Technology Co., Ltd.

Jul 2022 - Sep 2022

Algorithm Intern

Chengdu, Sichuan, China

- Researched the latest machine learning and industrial anomaly detection algorithms.
- Applied GAN models to reconstruct images to identify anomalies and evaluate the performance.
- Used the object detection neural network model to detect abnormal parts in high-speed rail supports.
- Assisted colleagues in developing a set of character recognition patents for LCD meters.

Selected Projects

KDEN-NSGA-II Scheduler for Flexible Job-Shop Scheduling

Jun 2025

- Developed a multi-objective FJSP solver based on enhanced NSGA-II with hybrid initialization and adaptive crossover, optimizing makespan, load balance, and machine idle time.
- Built a modular benchmarking framework supporting multiple baselines (NSGA-II, NSGA-III, SPEA2, multi-objective ACO) with KNN-based density selection for diversity-aware search.

High Performance Computing for Grey Wolf Optimizer (GWO) Optimization

Jan 2025

- Proposed HGT-GWO, integrating global historical best positions and trend guidance into GWO, outperforming standard GWO on 15 benchmark functions.
- Implemented a master-worker island parallelization scheme in C/MPI/OpenMP on UNITN's HPC cluster, reducing communication overhead through controlled synchronization.

Augmented Reality-Driven Robotic Arm Control for Industrial Automation

Sep 2024

- Built an AR-based robotic arm control system on Microsoft HoloLens 2, combining YOLOv8 and FastSAM for real-time gesture recognition and fingertip coordinate mapping.
- Supported multi-mode operation (gesture-based and interface-based) with robustness to hand occlusion in industrial environments.

Research on Semantic Segmentation Method of High-Resolution Remote Sensing Images

May 2023

Outstanding Undergraduate Graduation Project

- Developed an encoder-decoder model with residual-weighted attention for remote sensing segmentation, achieving F1-scores of 90.23% and 87.37% on ISPRS Potsdam and Vaihingen datasets.

Competitions

Kaggle March Machine Learning Mania 2025 - Silver Medal (64th / 1,727) <i>Solo</i>	Apr 2025
Kaggle BirdCLEF 2024 Competition - Bronze Medal (70th / 974) <i>Solo</i>	Jun 2024
Kaggle BirdCLEF+ 2025 Competition - Bronze Medal (199th / 2,025) <i>Solo</i>	Jun 2025
Mathematical Contest In Modeling (MCM), Finalist winner (top 1%) <i>Leader</i>	Apr 2021
• Designed a set of algorithms to assess the degree of hunger around the world and optimize the food supply chain. • Responsible for modeling, programming and part of paper writing. • The youngest winner in the history of Anhui University and the first winner of School of Internet.	
iCAN Innovation Contest 2021 (Anhui) - Second prize <i>Team member</i>	Oct 2021
• Developed a set of hardware system capable of intelligently monitoring and warning the abnormal behavior of the elderly. • Responsible for some STM32 development and project plan writing.	

Skills

- **Programming:** Python, C/C++, Java, Matlab, R, C#.
- **ML / DL Frameworks:** PyTorch, scikit-learn, TensorFlow, PaddlePaddle; MPI / OpenMP (HPC).
- **LLM / VLM:** multimodal LLMs (MLLMs), post-training (SFT, DPO, RL alignment incl. GRPO/GSPO), LoRA / PEFT, RAG, LLM agents.
- **Architectures:** Transformer, Mamba, CNN, RNN.
- **Areas:** Computer Vision, NLP, Reinforcement Learning, Mechanistic Interpretability, Evolutionary Algorithms.
- **Applications:** Image Classification & Segmentation, Object Tracking, Industrial Anomaly Detection, Time-Series Analysis.
- **Other:** Unity; Lean Manufacturing & Sustainable Management.

Language

- **English:** Fluent, TOEFL 102
- **Chinese:** Native Speaker